ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 1ST YEAR

ENGINEERING PHYSICS

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

BEng Mechatronics and Robotics Engineering, 1st year, semester 1 final examination: units and vectors, kinematics, Newton's laws, work, energy, power and momentum, statics and rotation, thermal physics and waves.

Written by Academic Central for practice, following first-year Engineering Physics syllabuses at South African universities (mechanics, thermal physics and waves for engineering students). It is not a university examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SIX questions. Answer ALL the questions.
2. Start each question on a new page. Show ALL your working, including formulae and substitutions.
3. Use g = 9.81 m/s² unless the question says otherwise.
4. Non-programmable scientific calculators may be used. Give answers correct to TWO decimal places, with SI units.
5. Number your answers exactly as the questions are numbered.

QUESTION 1: UNITS AND VECTORS

1.1 Write the units of force and of energy in terms of the SI base units (kg, m, s). (2)

1.2 Show that the equation $s = ut + \tfrac{1}{2}at^2$ is dimensionally correct. (2)

1.3 A force of 30 N acts due east and a force of 40 N acts due north on the same point. Calculate the magnitude and direction of the resultant force. (3)

1.4 Convert 72 km/h to m/s. (1)

1.5 Write 0.000 456 m in scientific notation and state the number of significant figures. (2)

[10]

QUESTION 2: KINEMATICS

2.1 A robot cart starts from rest and accelerates uniformly at 2.5 m/s² for 8 s. Calculate:

2.1.1 its final velocity (2)

2.1.2 the distance it travels in this time (2)

2.2 A ball is thrown vertically upwards at 19.6 m/s. Ignore air resistance. Calculate:

2.2.1 the time taken to reach the highest point (2)

2.2.2 the maximum height reached (2)

2.2.3 the total time until it returns to the hand (1)

2.3 A projectile is launched from level ground at 25 m/s at 30° above the horizontal. Calculate:

2.3.1 the time of flight (3)

2.3.2 the horizontal range (2)

2.3.3 the maximum height (2)

2.4 On a velocity–time graph, what does the gradient represent, and what does the area under the graph represent? (2)

2.5 A motor shaft turns at 1 200 revolutions per minute. Calculate its angular velocity in rad/s. (2)

[20]

QUESTION 3: FORCES AND NEWTON'S LAWS

3.1 State Newton's second law of motion in words. (2)

3.2 A 10 kg block is released on a rough slope inclined at 30° to the horizontal, as shown in FIGURE 1. The coefficient of kinetic friction is 0.2.

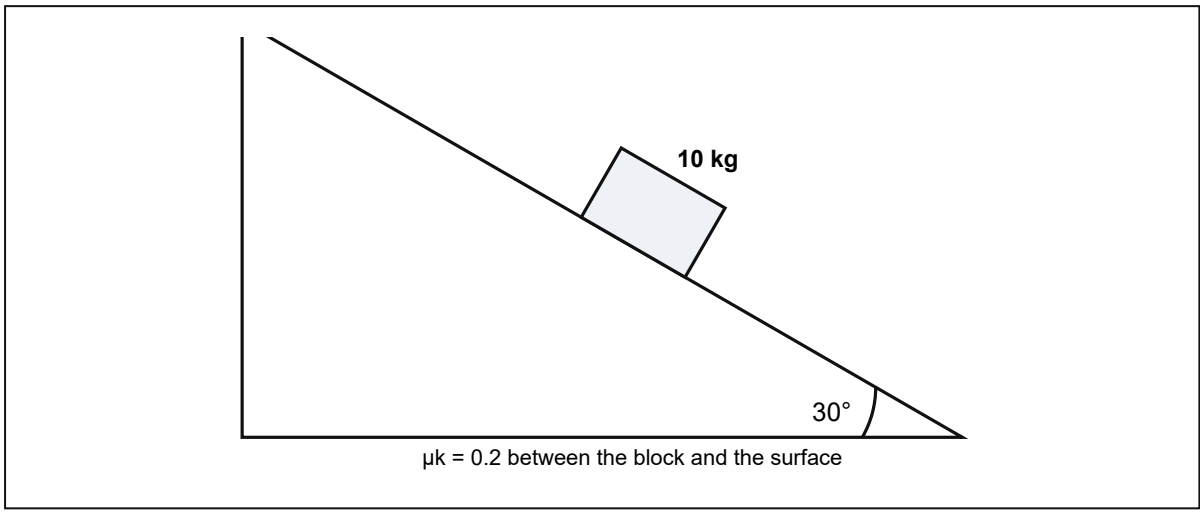


FIGURE 1: Block on an inclined plane

3.2.1 Draw a labelled free-body diagram of the block. (3)

3.2.2 Calculate the normal force on the block. (2)

3.2.3 Calculate the kinetic friction force. (1)

3.2.4 Calculate the acceleration of the block down the slope. (3)

3.3 A lift of mass 800 kg accelerates upwards at 1.5 m/s². Calculate the tension in the cable. (3)

3.4 Two masses, 5 kg and 3 kg, hang on either side of a light, frictionless pulley and are released from rest. Calculate:

3.4.1 the acceleration of the masses (3)

3.4.2 the tension in the string (3)

[20]

QUESTION 4: WORK, ENERGY, POWER AND MOMENTUM

4.1 Define work and give its SI unit. (2)

4.2 A 1 200 kg car moving at 20 m/s brakes to a stop over a distance of 50 m. Calculate:

4.2.1 the kinetic energy of the car before braking (2)

4.2.2 the average braking force (2)

4.3 A pump lifts 500 kg of water through a height of 12 m in 60 s.

4.3.1 Calculate the work done on the water. (2)

4.3.2 Calculate the useful output power. (1)

4.3.3 If the pump is 75% efficient, calculate its input power. (2)

4.4 A 2 kg trolley moving at 3 m/s collides with a stationary 1 kg trolley and they move off together.

4.4.1 Calculate their common velocity after the collision. (3)

4.4.2 Calculate the kinetic energy lost and say whether the collision is elastic or inelastic. (3)

4.5 A 0.15 kg ball hits a wall at 20 m/s and rebounds at 15 m/s. The contact time is 0.01 s. Calculate the change in momentum and the average force of the wall on the ball. (3)

[20]

QUESTION 5: STATICS AND ROTATION

5.1 A uniform beam AB is 4 m long and has a mass of 20 kg. It rests horizontally on supports at A and B. A load of 50 kg hangs 1 m from A. Calculate the reaction forces at A and B. (5)

5.2 A mechanic applies a 40 N force at right angles to the end of a 0.3 m spanner. Calculate the torque. (1)

5.3 State the two conditions for a body to be in equilibrium. (2)

5.4 A flywheel with a moment of inertia of 2 kg·m² speeds up uniformly from rest to 100 rad/s in 10 s. Calculate:

5.4.1 the angular acceleration (1)

5.4.2 the torque needed (2)

5.4.3 the rotational kinetic energy at 100 rad/s (2)

5.4.4 the angle turned through in the 10 s (2)

[15]

QUESTION 6: THERMAL PHYSICS AND WAVES

6.1 Calculate the heat needed to raise the temperature of 2 kg of water from 20 °C to 80 °C. ($c = 4\,186$ J/(kg·°C)) (2)

6.2 A steel rail is 25 m long. Calculate how much longer it gets when the temperature rises by 40 °C. ($\alpha = 12 \times 10^{-6}$ /°C) (2)

6.3 A gas occupies 0.02 m³ at 100 kPa and 300 K. It is compressed to 0.005 m³ and its temperature rises to 450 K. Calculate the new pressure. (3)

6.4 A sound wave has a frequency of 500 Hz and a wavelength of 0.68 m. Calculate its speed and its period. (3)

6.5 A 0.5 kg mass hangs from a spring with a spring constant of 200 N/m and oscillates in simple harmonic motion. Calculate the angular frequency, the frequency and the period. (3)

6.6 Explain what resonance is, and give one engineering example where it must be avoided. (2)

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TOTAL: 100

END OF QUESTION PAPER